Build Release Documents:-

- Once UNIT level testing is approved, developers will integrate the units and they will release build for testing in the form of URL's along with "Build Release Note".
- The Development Lead (or) Tech Lead will update the Build and Release Note into Testing Server and he will send an email to Test Lead/Testing team saying that
  - ✓ Build is ready for testing
  - ✓ Build URL's
  - ✓ He will mention Build Period

First Build Release Note contains:-

- Requirements implemented in this Build (Req#)
- URL of Build (Uniform Resource Locator)
- Admin Username and Password
- Database Details(HostName, PortNumber,Service Name, UserName,Password).

Second Build Release Note contains:-

- Defects fixed in this Build (Defect#)
- Requirements implemented in this Build (Req#)
- URL of Build (Uniform Resource Locator)
- Admin Username and Password
- Database Details(HostName, PortNumber,Service Name, UserName,Password).

Smoke Testing:-

Smoke Testing comes into the picture at the time of receiving build software from the development team.
The purpose of smoke testing is to determine whether the build software is testable or not.
It is done at the time of "building software." This process is also known as "Day 0".
It is a time-saving process. It reduces testing time because testing is done only when the key features of the application are not working or if the key bugs are not fixed. The focus of Smoke Testing is on the workflow of the core and primary functions of the application.
Testing the basic & critical feature of an application before doing one round of deep, rigorous testing (before checking all possible positive and negative values) is known as smoke testing
.

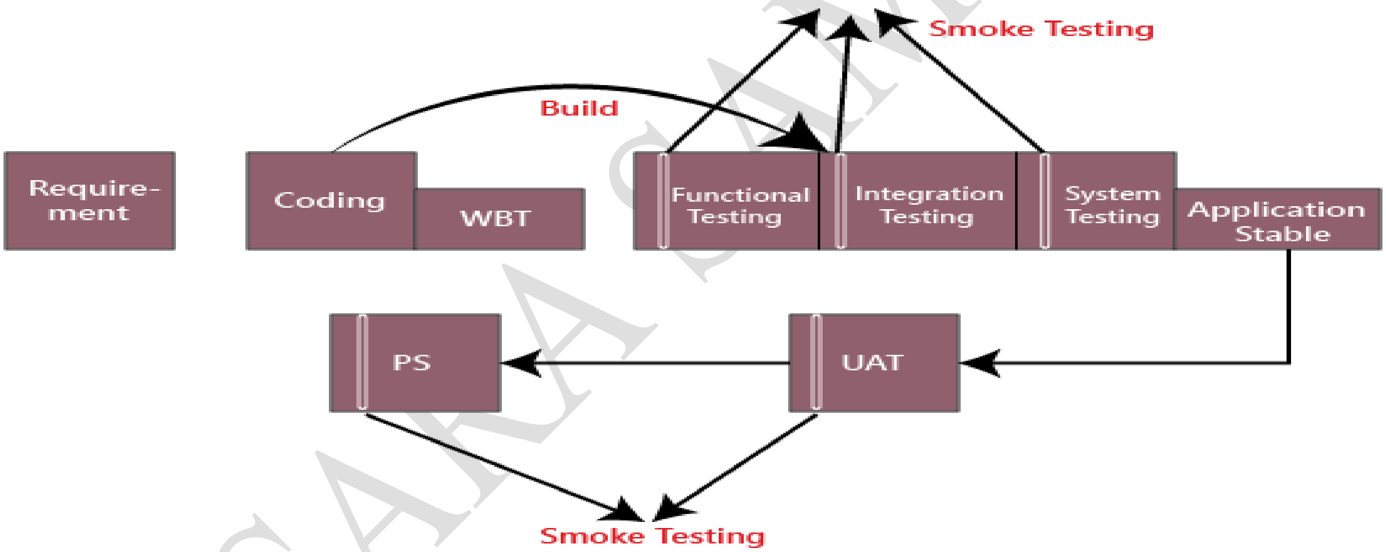
- In the smoke testing, we only focus on the positive flow of the application and enter only valid data, not the invalid data. In smoke testing, we verify every build is testable or not; hence it is also known as **Build Verification Testing.**
- When we perform smoke testing, we can identify the blocker bug at the early stage so that the test engineer won't sit idle, or they can proceed further and test the **independent testable modules.**

Process to conduct Smoke Testing:-

- Smoke testing does not require to design test cases. There's need only to pick the required test cases from already designed test cases.
- Smoke Testing focuses on the workflow of core applications so ,we choose test case suits that covers the major functionality of the application. The number of test cases should be minimized as much as possible and time of execution must not be more than half an hour.

When Smoke Testing Performed?

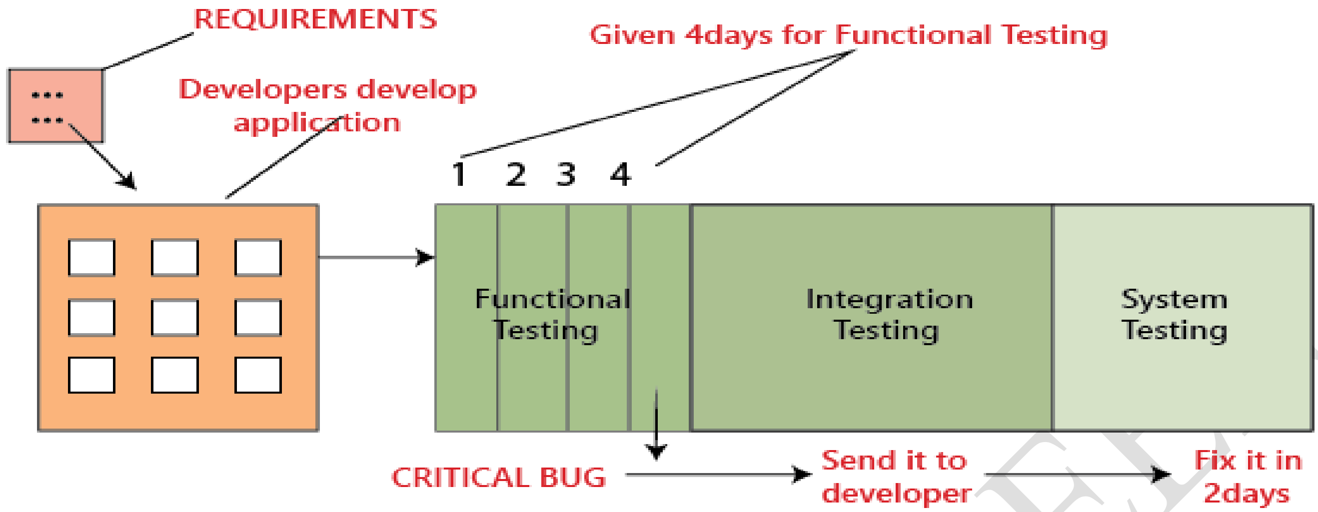
- Generally, whenever the new build is installed, we will perform one round of smoke testing because in the latest build, we may encounter the blocker bug. After all, there might be some change that might have broken a major feature (fixing the bug of or adding a new feature could have affected a major portion of the original software), or we do smoke testing where the installation is happening.
- When the stable build is installed anywhere (Test Server, Production Server, and User Acceptance testing), we do smoke testing to find the blocker bug.



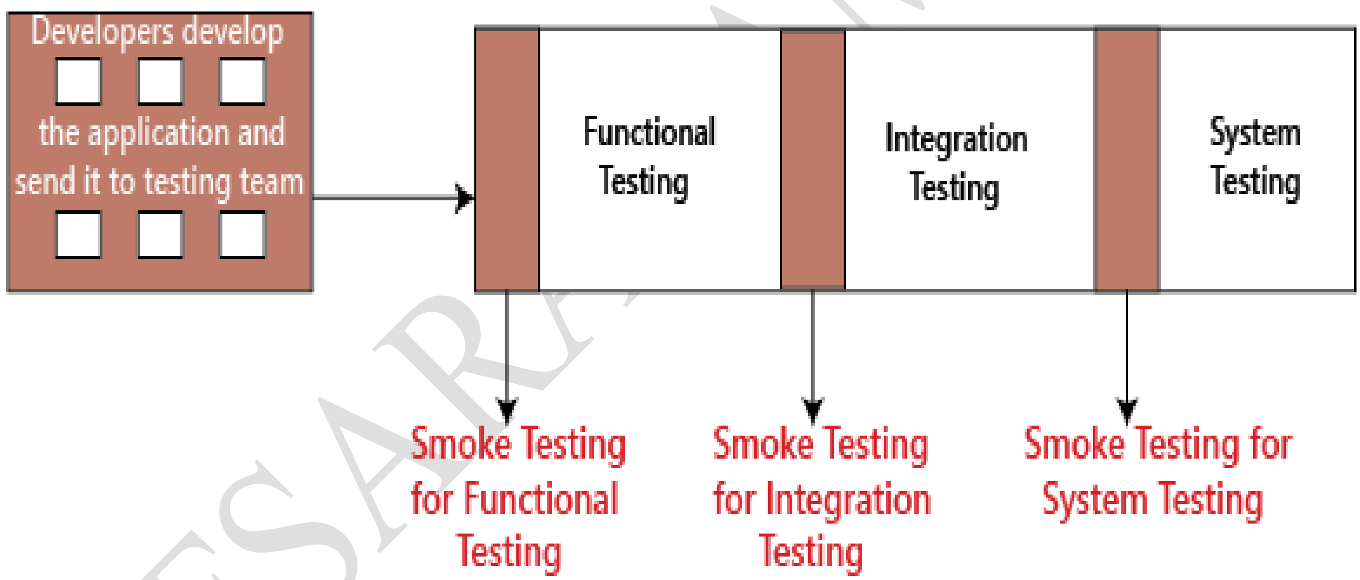
Different scenarios to do smoke testing:-

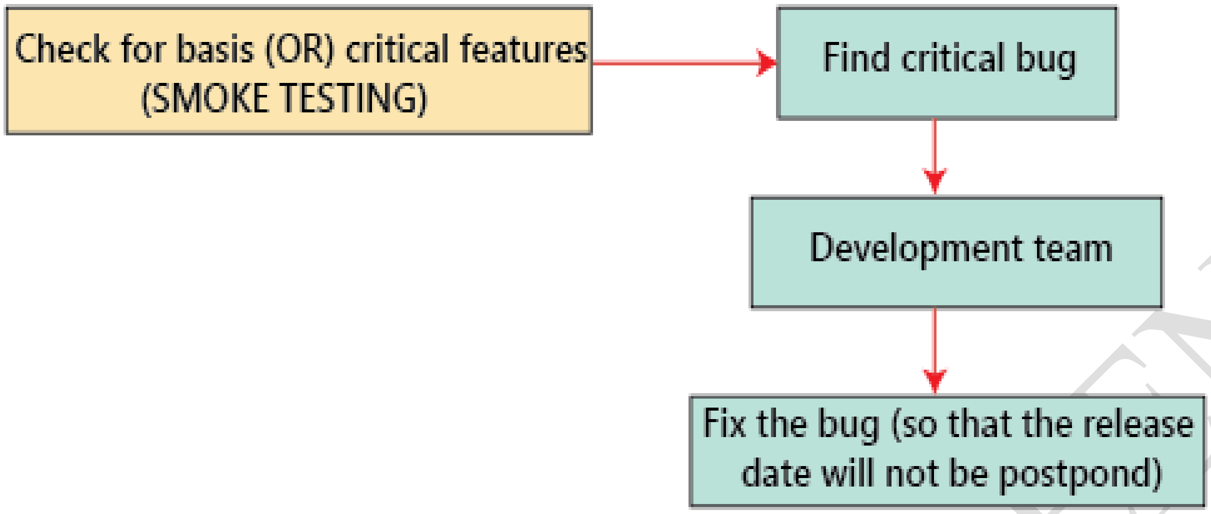
Scenarios 1:-

- The developer develops the application and handed over to the testing team, and the testing team will start the functional testing
- Suppose we assume that four days we are given to the functional testing. On the first day, we check one module, and on the second day, we will go for another module. And on the fourth day, we find a critical bug when it is given it to the developer; he/she says it will take another two days to fix it. Then we have to postpone the release date for these extra two days.



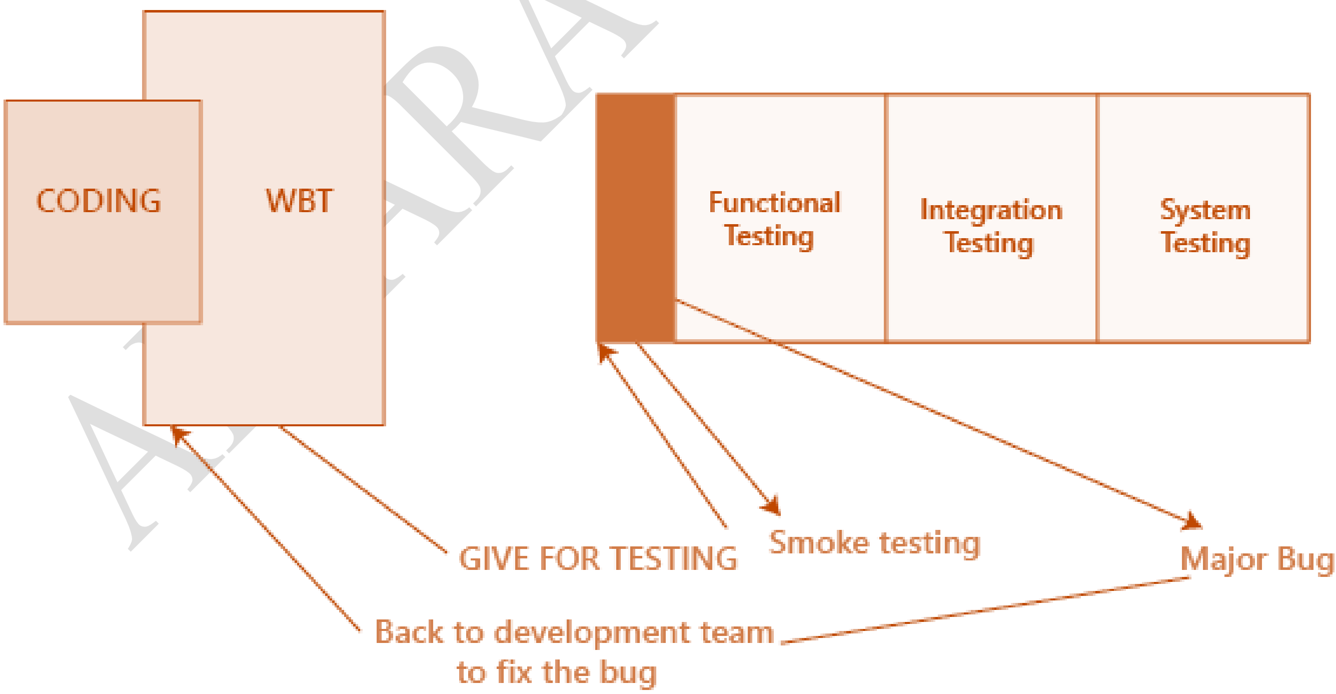
- To overcome this problem, we perform smoke testing, instead of the testing module by module thoroughly and come up with critical bug at the end, it is better to do smoke testing before we go for functional, integration and system testing that is, in each module we have to test for essential or critical features, and then proceed for further testing





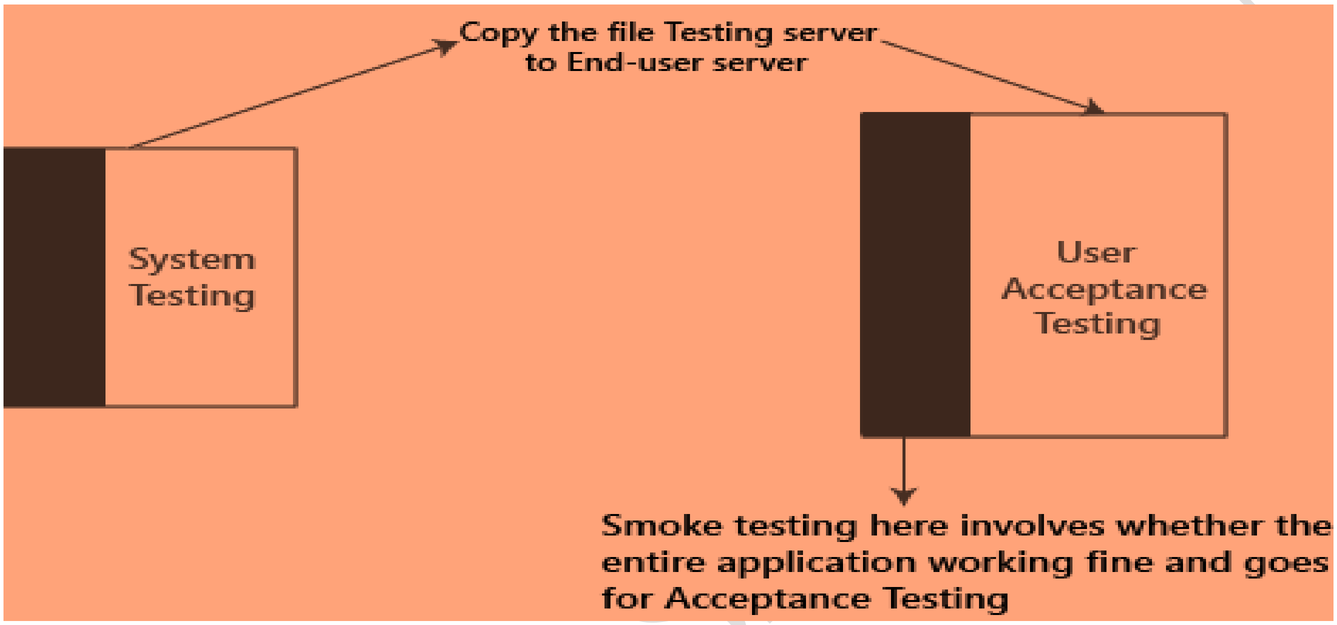
Scenario 2

- While performing the functional testing, if the test engineer identifies the major bug in the early stages, sometimes it is not suitable for the developer to find a major bug in the initial stages. So the test engineer will perform the smoke testing before doing the functional, integration, system, and other types of testing.
- While doing smoke testing, the test engineer finds the major bug; he/she will give to the development team for fixing the bug. After the bug is fixed, the test engineer will continue for further testing



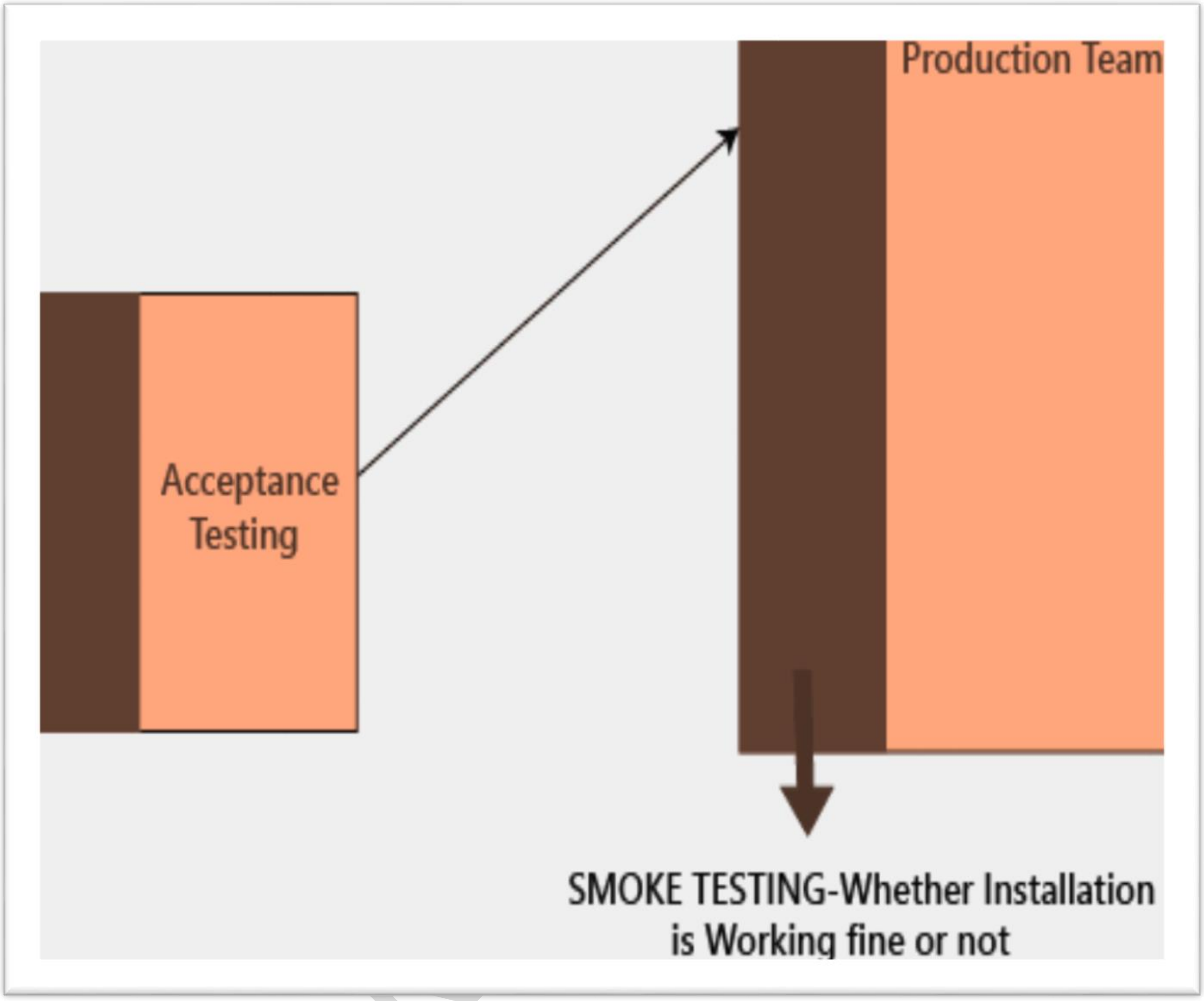
Scenario 3:-

- In this scenario, if we are already perform the smoke testing and found the blocker bug and also resolved that bug. After performing the system testing, we will send the application from the testing server to the end-user server for one round of user acceptance testing. And when the customer performs the acceptance testing and does not find any issues and satisfied with the application because we already perform the smoke testing.



Scenarios 4:-

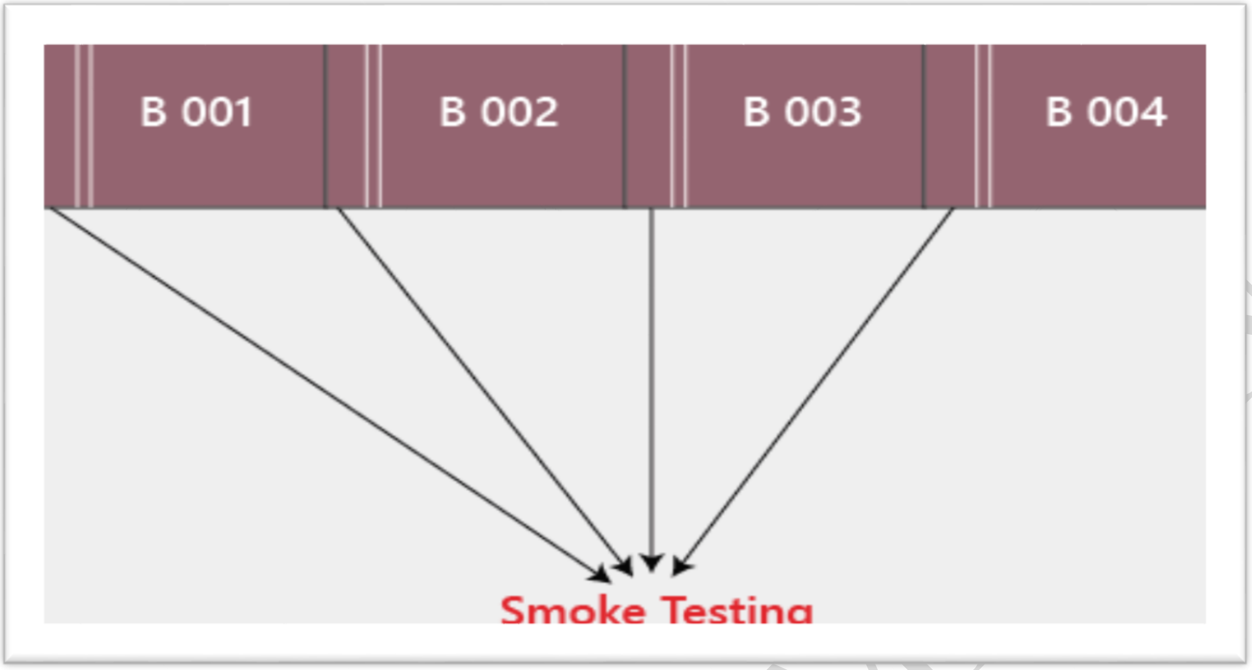
- Once the acceptance testing is done, the application will be deployed to the production server. We have done a round of smoke testing on the production server to check whether the application is installed correctly or not. If any real end-user will found any blocker bug, they will get irritated and will not use the application again, which may lead to the loss of customer business
- For not getting this problem in the future, the development team manager, the testing team manager, will take the customer login and do one round of smoke testing.



- In the production server, smoke testing can be done by the Business analyst (BA), Development team manager, testing team manager, build team, and the customer.

Why we do smoke testing?

- We will do the smoke testing to ensure that the product is testable.
- We will perform smoke testing in the beginning and detect the bugs in the basic features and send it to the development team so that the development team will have enough time to fix the bugs.
- We do smoke testing to make sure that the application is installed correctly.



Types of smoke testing:-

Formal smoke testing:-

- In this, the development team sends the application to the Test Lead. Then the test lead will instruct the testing team to do smoke testing and send the reports after performing the smoke testing. Once the testing team is done with smoke testing, they will send the smoke testing report to the test lead.

Informal smoke testing:-

- Here, the Test lead says that the application is ready for further testing. The test leads do not specify to do smoke testing, but still, the testing team starts testing the application by doing smoke testing.

Advantages of smoke testing:-

- It is a time-saving process.
- In the early stage, we can find the bugs.
- It will help to recover the quality of the system, which decreases the risk.
- It is easy testing to perform because it saves our test effort and time.